

Report VII

Nonlinear equation - finding the roots of polynomial

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1. Introduction

In many numerical problems, it is necessary to determine the roots of a nonlinear equation, i.e. values of x for which:

$$f(x) = 0$$

In this project, the following polynomial was analyzed:

$$f(x) = (x - 1)(x - 3)^6$$

This function has two real roots:

- $x = 1$, root of odd multiplicity
- $x = 3$, root of even multiplicity (degree $n = 6$)

The goal of the exercise is to compare three numerical methods used for solving nonlinear equations:

- Bisection method
- Secant method
- Newton-Raphson method

These methods differ in convergence speed, stability, and sensitivity to initial conditions. The polynomial serves as a good test case because it includes both a simple root and a multiple root.

2. Description of the Numerical Task

The task consisted of implementing and comparing iterative methods for finding roots of the given polynomial.

The following steps were performed:

- implementation of three algorithms:
 - Bisection
 - Secant
 - Newton-Raphson
- iterative computation of the root $x = 1$ using:
 - $it_{max} = 30$ and $\varepsilon = 10^{-12}$
 - starting points:
 - Bisection: $x_a = 0.1, x_b = 1.5$
 - Secant: $x_0 = 0.1, x_1 = 0.5$
 - Newton: $x = 0.1$
- saving iteration data:
 - iteration index k
 - approximation x_k
 - difference $\Delta_k = |x_k - x_{k-1}|$
- repeating calculations for root $x = 3$:
 - only Secant and Newton methods
 - $it_{max} = 300$
 - starting points:
 - Secant: $x_0 = 4.0, x_1 = 6.0$
 - Newton: $x = 6.0$
- visualization of:
 - convergence of x_k
 - convergence of Δ_k

3. Numerical Method

Bisection method

The method is based on interval halving. At each step, the interval containing the root is reduced by half. It is very stable but converges slowly (linear convergence).

Secant method

The method approximates the derivative numerically using two previous points. It converges faster than bisection but may be less stable.

Newton-Raphson method

The method uses the derivative which was calculated analytically:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

It has quadratic convergence near simple roots, making it the fastest method in most of the cases.

4. Implementation

The program was written in C++ and does not use any numerical libraries.

Main implementation steps:

1. Definition of function $f(x)$ and its derivative $f'(x)$
2. Implementation of three iterative methods following the provided pseudocode
3. Iterative loop with stopping condition:
 - maximum number of iterations
 - tolerance of ε

4. Saving results to files:

- iteration index
- current approximation
- difference between iterations

5. Generation of data for plotting convergence behavior

5. Results

Root $x = 1$

The plots show convergence of all three methods.

Figure 1. Approximated root position x_k for $x = 1$.

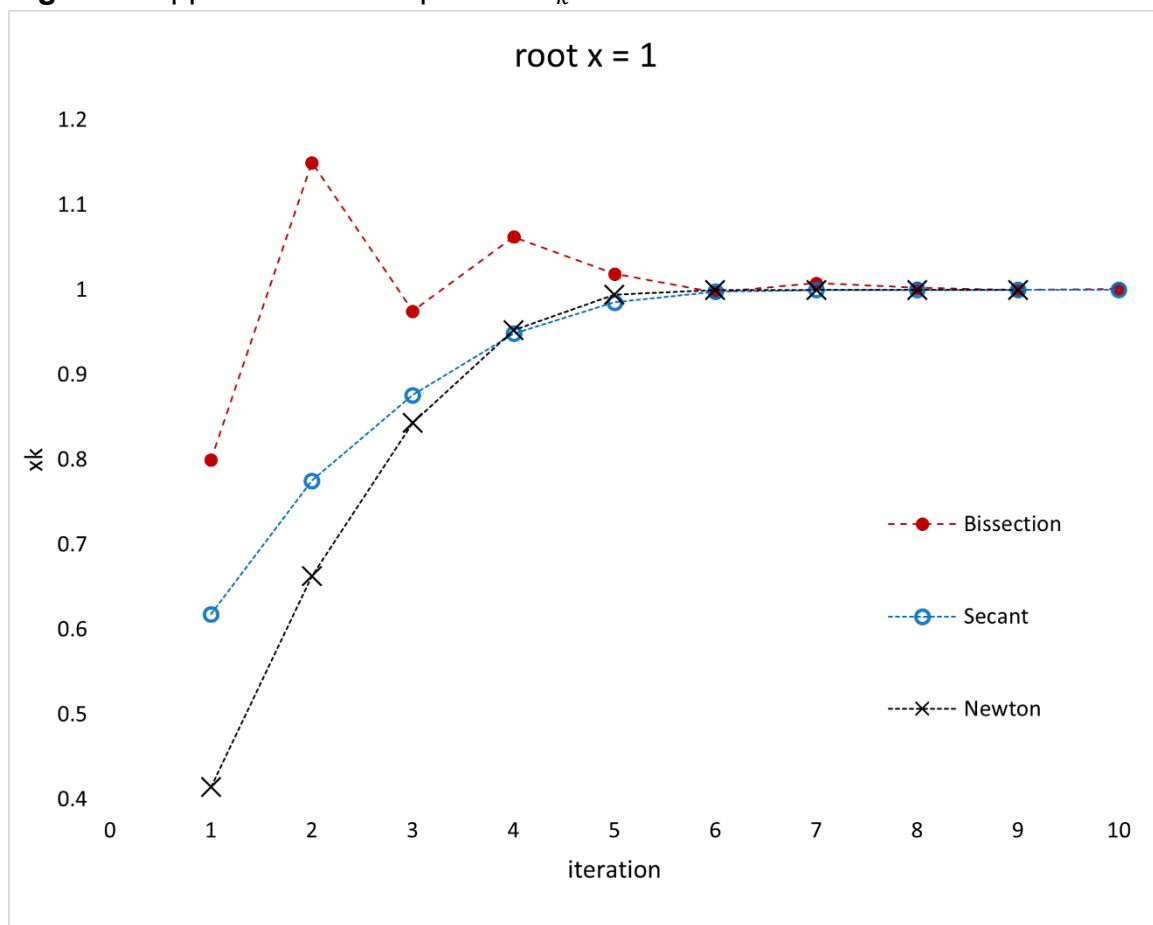
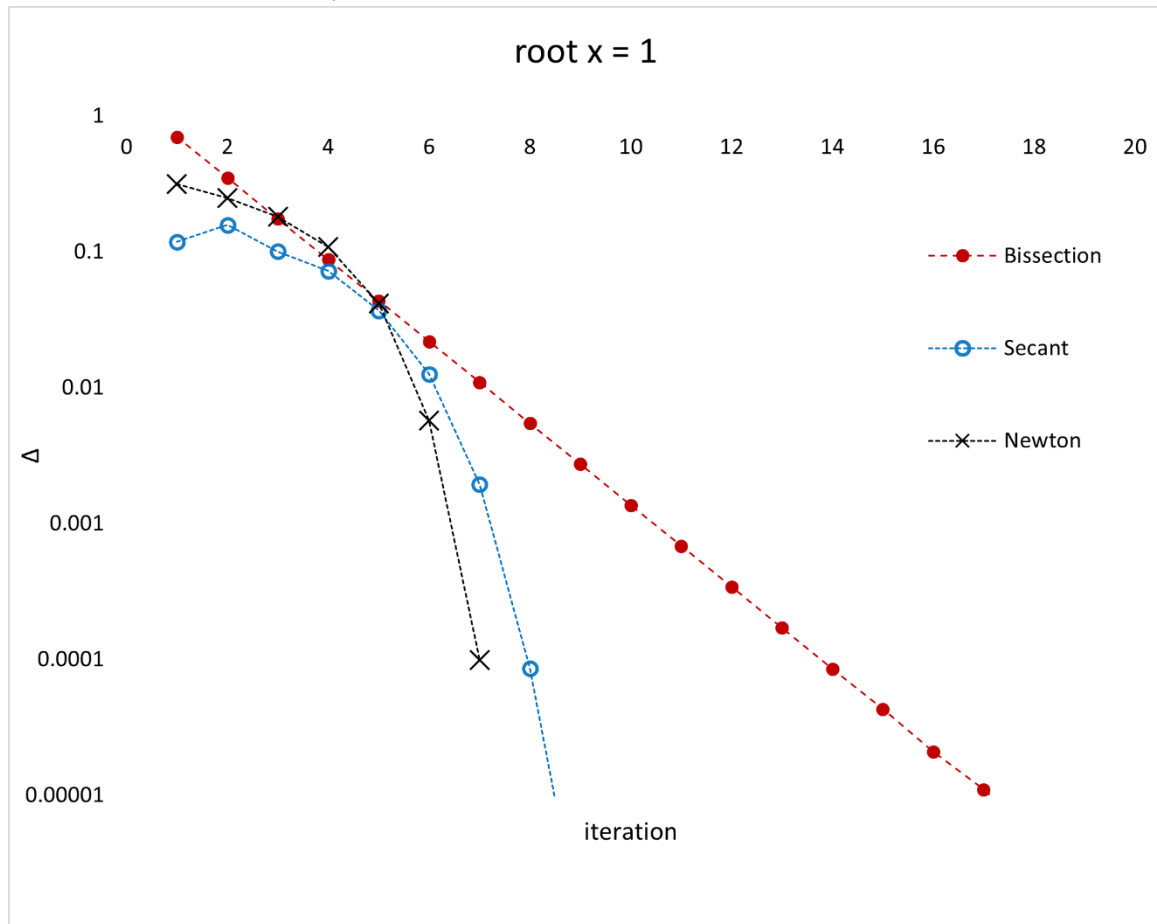


Figure 2. Difference Δ_k between two consecutive iterations for $x = 1$.



Observations:

- Bisection method converges steadily but slowly
- Secant method converges faster
- Newton-Raphson method converges the fastest (only a few iterations)

Root $x = 3$

Only Secant and Newton-Raphson methods were applied.

Figure 3. Approximated root position x_k for $x = 3$.

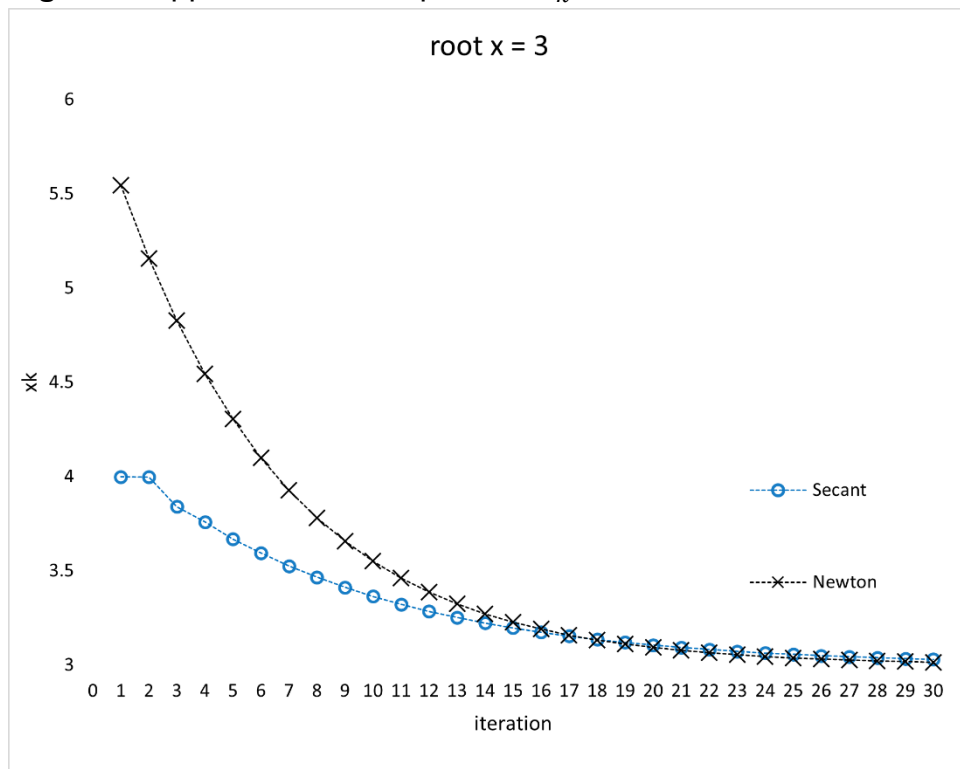
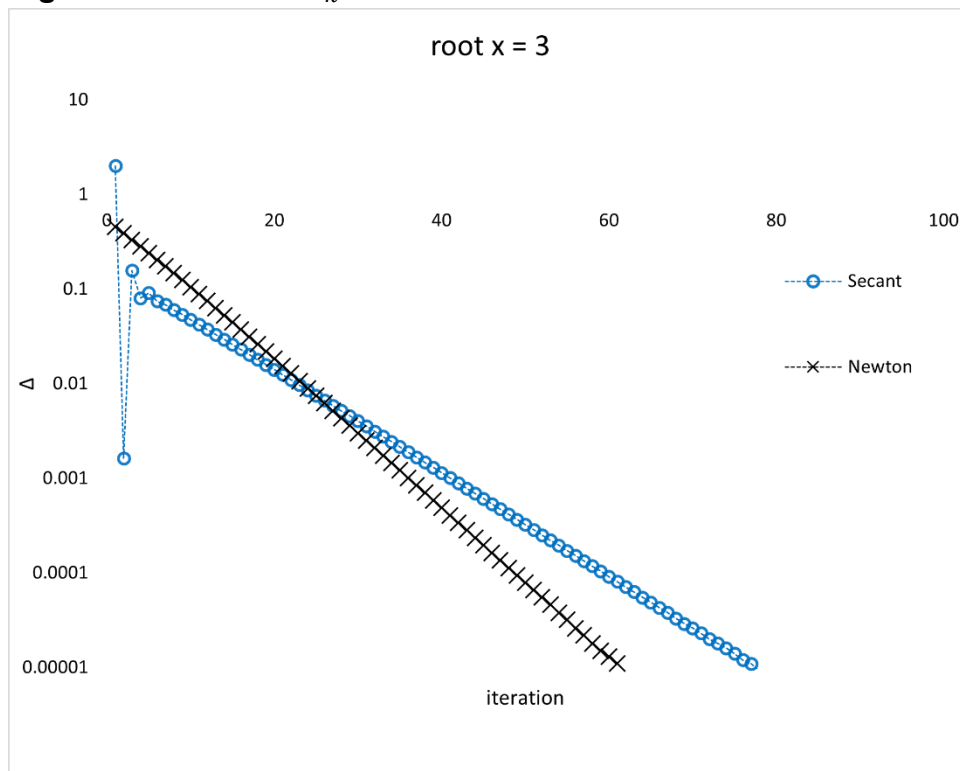


Figure 4. Difference Δ_k between two consecutive iterations for $x = 3$.



Observations:

- convergence is significantly slower than for $x = 1$
- both methods require many more iterations
- convergence curves are much less steep

This behavior is clearly visible also in the *plots presented on page 4 of the instruction*, where convergence for $x = 3$ is much slower compared to $x = 1$

6. Conclusions

Using the modified formula:

$$x_{k+1} = x_k - n \frac{f(x_k)}{f'(x_k)}$$

restores fast convergence for multiple roots.

In our case ($n = 6$):

- convergence becomes much faster
- number of iterations is greatly reduced

The following conclusions can be drawn overall:

- Newton-Raphson method is the most efficient for simple roots
- Secant method provides a good compromise between speed and simplicity
- Bisection method is the most stable but at the same time the slowest one
- multiple roots significantly reduce convergence speed
- modified Newton method effectively solves the problem of slow convergence
- proper choice of initial values strongly affects the performance

Overall, the study confirms that the choice of numerical method should depend on the properties of the function and the type of root.